1st Sessional Examination

B.Tech 2nd Semester

Course : Chemistry | Total Marks – 30 | Time – 1 h

1. Define enthalpy and internal energy. 10 moles of an ideal gas at the initial pressure of 1 atm at 0 degree C were expanded reversibly under isothermal conditions to a final pressure of 0.1 atmosphere. Calculate the change in internal energy, enthalpy, the work done by the gas and the heat absorbed by the system ($R=8.314\ JK^{-1}mol^{-1}$). (2+5)
2. What do you mean by entropy. Calculate the increase in entropy in the evaporation of one mole of water at 373 K. Latent heat of vaporization of water in 2.26 kJ/g (molar mass water 18g/mol). (3)

3. Determine the concentration of a solution of thickness 3 cm that transmits 30% incident light. Given Extinction coefficient, € = 4000 $dm^3mol^{-1}cm^{-1}$. (4)

4. Identify the possible electronic transitions that can take place in the given molecules when it absorbs UV-Visible radiation

CH_3CHO, CH_3-CH_3 (2)

5. Write the statement of Beers Law. (2)

6. Write any two applications of IR Spectroscopy. (2)

7. What is corrosion? Explain dry and wet corrosion. (1+3)

8. Discuss the following

(a) Differential aeration corrosion (b) Waterline corrosion (3+3)